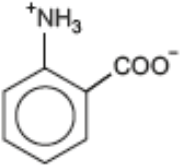


Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)	(i)		sulfur oxide dichloride / thionyl chloride / SOCl_2 phosphorus(III) chloride / PCl_3 phosphorus(V) chloride PCl_5	1			1		1
		(ii)		it does not absorb in the visible region	1			1		
		(iii)		loss of $191 - 91 / 100$ (1) could be $(\text{C}_2\text{H}_5)_2\text{N}-\text{C}=\text{O}$ which has M_r of $29 + 29 + 14 + 28 = 100$ (1) accept $\text{C}_5\text{H}_{10}\text{NO}$			2	2		
	(b)	(i)		alkaline potassium manganate(VII) solution / $\text{MnO}_4^- + \text{OH}^-$	1			1		1
		(ii)		as the reaction is carried out in alkaline solution the salt of the acid rather than the acid itself is produced			1	1		1
	(c)	(i)				1		1		
		(ii)		the ester contains a basic $-\text{NH}_2$ group which can accept a proton		1		1		
		(iii)	I	water is produced as during the esterification reaction / an aqueous solution of sodium carbonate has been added		1		1		1
			II	electrically heated / hot water bath			1	1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		2.0×10^{-5}		1		1	1	
		(ii)		as BA is produced it dissolves preferentially in the hexane (1) this removes BA from the reaction and moves the position of equilibrium to the right (1) 2-aminobenzoic acid remains largely in the aqueous alcohol mixture (1)			3	3		3
				Question 9 total	3	4	7	14	1	8